

Sean Lee

Electrical Engineering Graduate

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Professional Experience

Research Assistant, *University of British Columbia*

05/2026 – 07/2026

Kelowna, BC, Canada

- Selected and sourced electronic components for an Arduino-based gas sensing system used in an alcohol recycling application.
- Integrated an MQ-3 gas sensor, relay module, and Arduino Uno to implement automated safety shutoff functionality.
- Performed electrical integration, wiring, debugging, and functional validation of prototype hardware.
- Designed and fabricated custom Fusion 360 enclosures and electrical junction boxes using FDM 3D printing.

Research Engineering Work Study, *University of British Columbia*

05/2024 – 01/2025

Kelowna, BC, Canada

- Designed and prototyped a cleanroom-compatible spin coating system for laboratory use.
- Developed mechanical components using Fusion360 and FDM printing.
- Integrated an Arduino Mega control system with brushless DC motor control hardware for closed-loop rotational speed control.

Projects

MEMS Signal Generator

- Designed MEMS test structures incorporating cantilevers, serpentine springs, and conductive contacts using KLayout.
- Characterized device performance using electrical continuity and mechanical compliance testing.
- Investigated failure mechanisms including stiction, anchor damage, and contact misalignment through root-cause analysis.
- Iteratively refined device designs to improve mechanical reliability, manufacturability, and fabrication yield.

Automated Liquid Handling System

- Designed an automated fluid handling platform using ESP32-based embedded control.
- Integrated diaphragm pumps, valves, and custom SLA-fabricated fluid junctions to enable intake, storage, and dispensing of liquids.
- Developed code to validate functions and integrated proprietary software for real-time control.
- Demonstrated consistent liquid transfer of 5mL.

Research Experience

Microfabrication of MEMS using Electroplating,

06/2024 – 07/2024

Course Project- Microfabrication Technologies

- Developed an electroplating-based MEMS fabrication process through literature review, process planning, and experimental validation.
- Independently fabricated MEMS test structures using photolithography, thin-film deposition, and electroplating in a Class-100 cleanroom environment.
- Documented process parameters, fabrication challenges, and device performance to support process reproducibility.

- A 1.3 micrometer layer of copper was targeted and an error of approximately 10% was achieved

Parylene Bonding for Microfluidic Chips, Undergraduate Lab Assistant

09/2023 – 08/2025

- Developed and optimized a parylene-C bonding process for microfluidic device fabrication with a success rate of 93%.
- Designed test structures using Fusion360 and fabricated samples using SLA printing, parylene deposition, and thermal compression bonding.
- Performed process characterization by systematically varying film thickness, and bonding temperature to identify optimal process parameters across 50 tests.
- Conducted mechanical bond testing and surface characterization to evaluate bond integrity and repeatability.

Skills

- Hardware & Test Equipment** — Oscilloscope | Digital Multimeter | Power Supplies | Function Generator | Arduino | ESP32
- CAD** — Fusion 360 | KLayout
- Programming Languages** — MATLAB | Python | C++ | Verilog
- Design and Simulation** — Cadence Virtuoso | AWR Design Environment | LTSPICE
- Process Engineering** — Process Optimization | Experimental Design | Process Characterization | Failure Analysis | Device Fabrication | Thin Film Deposition

Leadership and Communication

- Go Global Program Ambassador**

08/2025 – 04/2026

Mentored students pursuing international exchange opportunities.
- Undergraduate Research Presenter**

01/2025

Presented research findings to interdisciplinary audiences.
- Okanagan Tech Industry Night Guest Speaker**

02/2024

Presented research findings to university students, faculty and industry members

Languages

- English**
 - Native Level
- Korean**
 - Conversational

Education

- B.A.Sc. Electrical Engineering, University of British Columbia**

09/2022 – 06/2026

Relevant courses:

Kelowna, BC, Canada

 - Microelectromechanical Systems
 - Microfabrication Technology
- Exchange Student, Korea University**

03/2025 – 06/2025

Seoul, South Korea